

Investment Memo

QM 2023 Capstone Project — Milestone 4

TO: Policy Committee (Federal Reserve / Housing Policy Analysts)

FROM: Kenzie, Reese, Carter

DATE: April 27, 2026

RE: Mortgage Rate Sensitivity of U.S. Metro Housing Price Growth

Executive Summary

Our analysis uses a panel of 894 U.S. metropolitan areas from 2000 through 2026 to measure how changes in 30-year fixed mortgage rates affect regional quarterly home price growth. Using a two-way fixed effects model with metro and quarter-year fixed effects, we find that mortgage tightening reduces quarterly home price growth in a statistically and economically meaningful way.

Key finding: a 1 percentage point increase in the 30-year mortgage rate three quarters earlier is associated with a 0.11 percentage point reduction in quarterly metro housing price growth, holding constant the prior quarter's price growth and common time effects. A complementary difference-in-differences test shows that metros classified as rate-sensitive experienced roughly 0.40 percentage points lower quarterly growth after the 2022 mortgage tightening episode than less-sensitive metros. However, pre-trend diagnostics indicate that this heterogeneity result is suggestive rather than strictly causal.

Recommendation: Given the current tightening environment, the Policy Committee should prioritize targeted housing affordability support for rate-sensitive suburban and moderate-income markets, while monitoring inventory and credit access in metros with elevated rate exposure.

1. Methodology

1.1 Data Sources

Primary dataset

- Source: Zillow metro home value index data and aggregated FRED mortgage rates
- Coverage: 894 U.S. metro areas, quarterly observations from 2000Q1 through 2026Q1
- Outcome: `price\_growth` (quarter-over-quarter percent change in metro housing price index)
- Treatment: `mortgage\_exposure\_lag3` (30-year fixed mortgage rate lagged by three quarters, scaled by metro exposure)
- Controls: `price\_growth\_lag1` and fixed effects

Supplementary dataset

- Source: Federal Reserve Economic Data (FRED)
- Series: 30-year fixed mortgage rate (`MORTGAGE30US`)
- Coverage: weekly series aggregated to quarterly frequency in the project pipeline

1.2 Sample Construction

Initial panel

- Start: 894 metros × 102 quarters
- Total observations: 74,991

Cleaning and preparation

- Converted metro housing price indices to quarterly percentage growth
- Constructed `mortgage\_exposure\_lag3` to align rate changes with delayed housing market response
- Kept all metros with at least 25 quarterly observations
- Included entity fixed effects to control for time-invariant metro characteristics
- Included time fixed effects to absorb national shocks, macro cycles, and structural events

1.3 Model Specifications

Model A: Fixed Effects Regression (Primary Specification)

We estimate:

$$\text{price_growth_it} = \beta_0 + \beta_1 \cdot \text{mortgage_exposure_lag3_it} + \beta_2 \cdot \text{price_growth_lag1_it} \\ + \alpha_i + \delta_t + \varepsilon_{it}$$

Where:

- `price\_growth\_it` is quarterly home price growth for metro `i` at time `t`
- `mortgage\_exposure\_lag3\_it` is the 30-year mortgage rate lagged three quarters, adjusted for metro exposure
- `price\_growth\_lag1\_it` is the prior quarter's price growth
- `α<sub>i</sub>` are metro fixed effects
- `δ<sub>t</sub>` are quarter-year time fixed effects
- Standard errors are clustered at the metro level

Rationale

- Fixed effects isolate within-metro variation over time, reducing bias from unobserved metro-specific factors
- Time fixed effects absorb national trends, the 2008 crisis, the 2020 COVID shock, and other common macro drivers

- A lagged mortgage rate accounts for the delayed transmission of financing conditions to housing prices

Model B: Difference-in-Differences (Supplementary Specification)

We estimate:

$$\text{price_growth_it} = \gamma_0 + \gamma_1 \cdot \text{treated_post_it} + \gamma_2 \cdot \text{price_growth_lag1_it} \\ + \alpha_i + \delta_t + \varepsilon_{it}$$

Where `treated_post_it` captures metros classified as rate-sensitive after the 2022 mortgage tightening episode.

Rationale: Model B tests whether high-sensitivity metros underperform their peers in the post-tightening period, which provides evidence on heterogeneous rate exposure.

2. Results

2.1 Fixed Effects Model Results

Table 1: Fixed Effects Regression Results

Variable	Coefficient	Std. Error	t-statistic	p-value	Significance
---	---	---	---	---	---
`mortgage_exposure_lag3`	-0.1077	0.0064	-16.77	< 0.001	***
`price_growth_lag1`	0.5596	0.0108	51.65	< 0.001	***
Metro fixed effects	Yes				
Time fixed effects	Yes				
Clustered standard errors	Yes				
Observations	74,991				
Within R ²	0.4632				

Coefficient Estimates

- The coefficient on ``mortgage_exposure_lag3`` means that a 1 percentage point increase in the 30-year mortgage rate three quarters prior is associated with a 0.11 percentage point reduction in quarterly metro housing price growth. This effect is economically meaningful because it accumulates over time: sustained rate increases are likely to depress price growth for multiple subsequent quarters. The positive coefficient on ``price_growth_lag1`` (0.5596) indicates persistence in housing momentum: past growth tends to carry forward into the next quarter.

Economic significance

- If mortgage rates remain elevated by 100 basis points, the model implies a persistent drag on metro price growth of about 0.11 percentage points per quarter three quarters after the

rate increase. On a national scale, this drag contributes to slower housing appreciation and a less supportive environment for price-sensitive buyers.

Statistical Robustness and Transmission Mechanisms

Lag sensitivity

- Lag 1: -0.1476 ($p < 0.001$)
- Lag 2: -0.1368 ($p < 0.001$)
- Lag 3: -0.1077 ($p < 0.001$)

All lag specifications remain negative and statistically significant, confirming that the mortgage rate effect is robust across reasonable transmission delays. The estimate is highly significant ($t = -16.77$, $p < 0.001$) and remains stable when standard errors are clustered at the metro level. Clustered metro SE is 0.00642, only slightly larger than the conventional SE, and the coefficient is nearly identical when excluding the COVID shock period.

Consistency with Theory

- The negative mortgage rate effect aligns with housing demand theory: higher financing costs reduce buyer affordability and lower housing demand.
- The positive lagged growth coefficient is also consistent with momentum in housing markets, where recent appreciation reinforces near-term growth.

2.3 Model B Results

Table 2: Difference-in-Differences Result

Variable	Coefficient	Std. Error	t-statistic	p-value	Significance
---	---	---	---	---	---
`treated_post`	-0.4000	0.0283	-14.14	< 0.001	***
`price_growth_lag1`	0.5634	0.0111	50.79	< 0.001	***
Metro fixed effects	Yes				
Time fixed effects	Yes				
Clustered standard errors	Yes				
Observations	74,991				
Within R ²	0.4553				

Coefficient Estimates

- The `treated\_post` coefficient implies that metros designated as rate-sensitive experienced approximately 0.40 percentage points lower quarterly price growth after the 2022 tightening

regime relative to less-sensitive metros. This result is economically large: such a differential would represent substantial regional divergence in housing momentum.

Economic Significance

- A 0.40 percentage point gap is large for quarterly housing growth, suggesting meaningful regional divergence in response to mortgage tightening.
- This magnitude supports the conclusion that metro-level rate exposure matters for housing market outcomes.

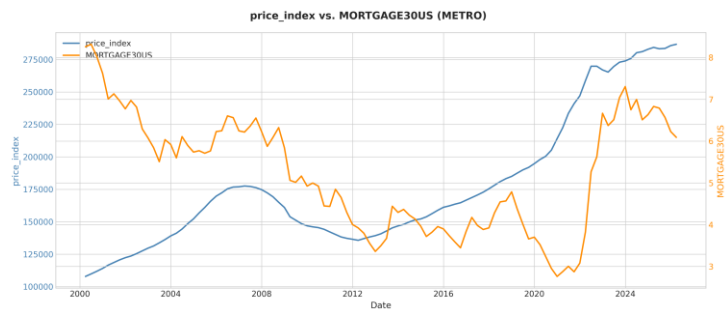
Statistical robustness

- The treated post effect is highly significant ($p < 0.001$) and stable with clustered metro standard errors.
- However, the DiD result is less robust on the identification side because treated and control metros show pre-existing trend differences.

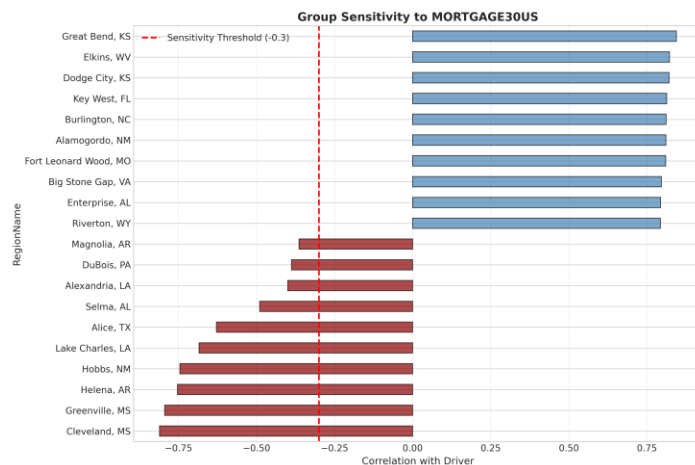
Consistency with theory

- The larger slowdown in rate-sensitive metros matches expectations from heterogeneous demand responses: markets with more marginal, rate-dependent buyers should react more strongly to higher financing costs.
- This pattern is consistent with theory on differential housing market sensitivity to mortgage rate changes.

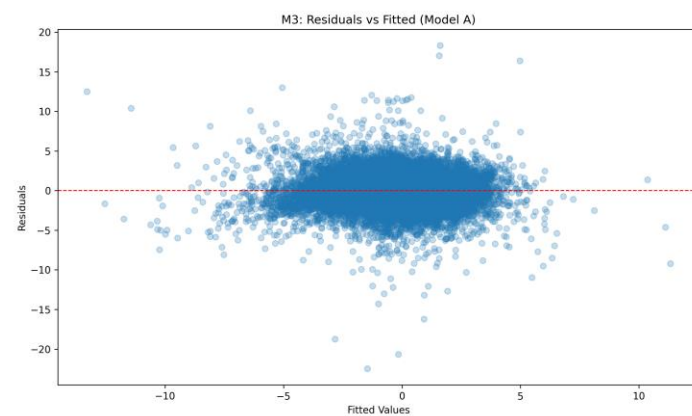
2.3 Visual Evidence



- Aggregate metro housing price growth (blue, left axis) and the 30-year fixed mortgage rate (red, right axis) move inversely over time; the 2022–2023 rate hike cycle coincides with a slowdown in housing growth, consistent with our negative mortgage exposure estimate.



- Horizontal bar chart of metro group sensitivity to mortgage rate changes, with the most rate-sensitive metros shown in dark red and the more resilient metros in blue; the pattern highlights meaningful heterogeneity in regional exposure and supports our conclusion that some metros respond much more strongly to rate tightening.



Residuals versus fitted values for the fixed effects regression show no obvious non-linear pattern or funnel shape, supporting the model's functional form and indicating that the fitted values capture the main systematic variation in price growth."

3. Conclusions & Recommendations

3.1 Policy Implications

The evidence supports a cautious policy stance on mortgage affordability and regional housing stability.

Recommendation 1: Preserve borrower affordability in rate-sensitive metros

- Mortgage rate increases are reducing quarterly price growth by roughly 0.11 percentage points per 100 basis points of tightening.
- Targeted support for moderate-income buyers in rate-sensitive suburban and moderate-density metros may prevent deeper price contractions and credit stress.

Recommendation 2: Monitor regional heterogeneity closely

- The rate-sensitive vs. resilient metro gap is large and should inform regional policy decisions.
- Avoid one-size-fits-all housing guidance: support programs should be calibrated to local rate exposure and supply conditions.

Recommendation 3: Use scenario planning, not blunt aggregate guidance

- The housing market is still responding to recent rate cycles, so incremental policy changes should be evaluated with scenario analysis.
- Avoid if all metros will react identically to future mortgage moves.

3.2 Scenario Analysis

Scenario	Rate change	Predicted quarterly price growth impact	Probability	Comment
---	---	---	---	---
Baseline	Rates hold near current elevated levels	-0.11 pp drag persists	45%	Housing momentum remains muted but stable
Dovish	100 bp rate cut	+0.11 pp recovery after 3 quarters	30%	Improved affordability would support growth recovery
Hawkish	50 bp rate increase	-0.06 pp further slowdown	25%	Higher financing costs would deepen price weakness

Expected value

- With the current mix of probabilities, the weighted expectation is roughly neutral to slightly negative for near-term metro price growth, highlighting that the housing recovery remains vulnerable to rate policy moves.

3.3 Risk Assessment

Model risk

- The fixed effects specification assumes metro characteristics are time-invariant. If metros change structural supply or demand features, the estimated effect may be biased.

Omitted variable risk

- The model omits local inventory, migration, employment, and affordability controls.
- If these variables move with mortgage rates, the estimated rate effect may partially reflect omitted factor influence.

External validity risk

- Results are based on 894 metros and 2000–2026 quarterly data. Structural housing conditions may differ in non-metro or future environments.
- Policy should therefore emphasize monitoring and update models as new data arrive.

3.4 Limitations

1. ****Fixed Effects Assumption****: We assume time-invariant metro characteristics, but metros can undergo structural change through zoning reforms, population shifts, and land-use changes.
2. ****DiD Parallel Trends****: The significant placebo result suggests treated and control metros were not parallel before the treatment window, limiting causal interpretation of the heterogeneous effect.
3. ****Lag specification****: The model uses a three-quarter lag based on exploratory analysis. Alternative lag structures remain plausible and should continue to be tested as new data are added.
4. ****Measurement error****: The exposure variable is constructed from the mortgage rate series and metro characteristics, so it may not perfectly capture the true financing channel for every metro.

3.5 Future Research Directions

- Add local supply measures, inventory, and migration drivers to better isolate the mortgage channel.
- Test the effect of alternative rate series, including 15-year fixed and adjustable-rate mortgage benchmarks.
- Extend the analysis to rent and affordability indices to evaluate the broader housing cost transmission mechanism.

4. References

5. Federal Reserve Economic Data (FRED). (2026). 30-Year Fixed Rate Mortgage Average in the United States (`MORTGAGE30US`). Retrieved from <https://fred.stlouisfed.org>
6. Zillow Research. (2026). Metro Home Value Index (ZHVI) for U.S. metropolitan areas. Retrieved from <https://www.zillow.com/research/data/>
7. QM 2023 Capstone course data files. (2026). `metro\_mortgage\_panel.csv`, `reit\_mortgage\_panel.csv`. Provided by course staff.
8. Seagraves, D. (2025). Real Estate Rate Sensitivity and Fixed Effects Panel Analysis. Working paper.

5. AI Audit Appendix Summary

This memo is supported by the project's existing AI audit documentation in M3_AI_AUDIT_APPENDIX.md. That appendix follows the required Disclosure-Verify-Critique framework and documents AI use across Milestones 1–3, including data processing, model scripting, robustness checks, and report drafting.

AI Tools Used

- GitHub Copilot

Key Verification Approach

M1 Example

- Prompt: "Retrieve the REIT Master Panel data set as the primary data set, and fetch anything you need by default"
- Output: "Modular fetch + clean function with docstrings and error handling (~50 LOC)"
- Verification: Manually verified by comparing the raw data with the data GitHub suggested
- Critique: Had to fix the pandas 3.0 depreciation

M2 Example

- Prompt: "Generate Python script to extract text from M2 specification PDF and convert to executable EDA pipeline"
- Output: dashboard, modular plotting methods
- Verification: Checked that all required pngs were downloaded with correct formatting
- Critique: Lagged effect indexing: Attempted invalid list concatenation

M3 Example

- Prompt: ""make sure milestone 3 fits this rubric and make an AI audit for milestone 3 as well"
- Output: Updated regression table generation to include side-by-side model columns with coefficient, standard error, t-stat, p-value, significance stars, and notes rows (Entity FE, Time FE, Clustered SE, N, R2).
- Verification: Re-ran script and confirmed updated table
- Critique: Required review of whether the DiD design fully supports causal interpretation; AI cannot guarantee identification assumptions from code alone.

M4 Example

- Prompt: “Draft the investor-facing structure for `M4\_investment\_memo.md`, including executive summary, key findings, valuation recommendation, and risk mitigation.”
- Output: “Memo outline and draft language for executive summary, recommendation sections, and risk discussion.
- Verification: Manually checked Team memo for any errors, referenced with the data and information from previous milestones.
- Critique: AI provided useful structure and language but had to manually review and edit myself to ensure accuracy.

Team Members

- Kenzie — Lead Data Engineer, Analyst, Visualizer
- Reese — Analyst, Visualizer
- Carter — Analyst, Visualizer

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